Time Series Anomaly Detection in Streaming Settings

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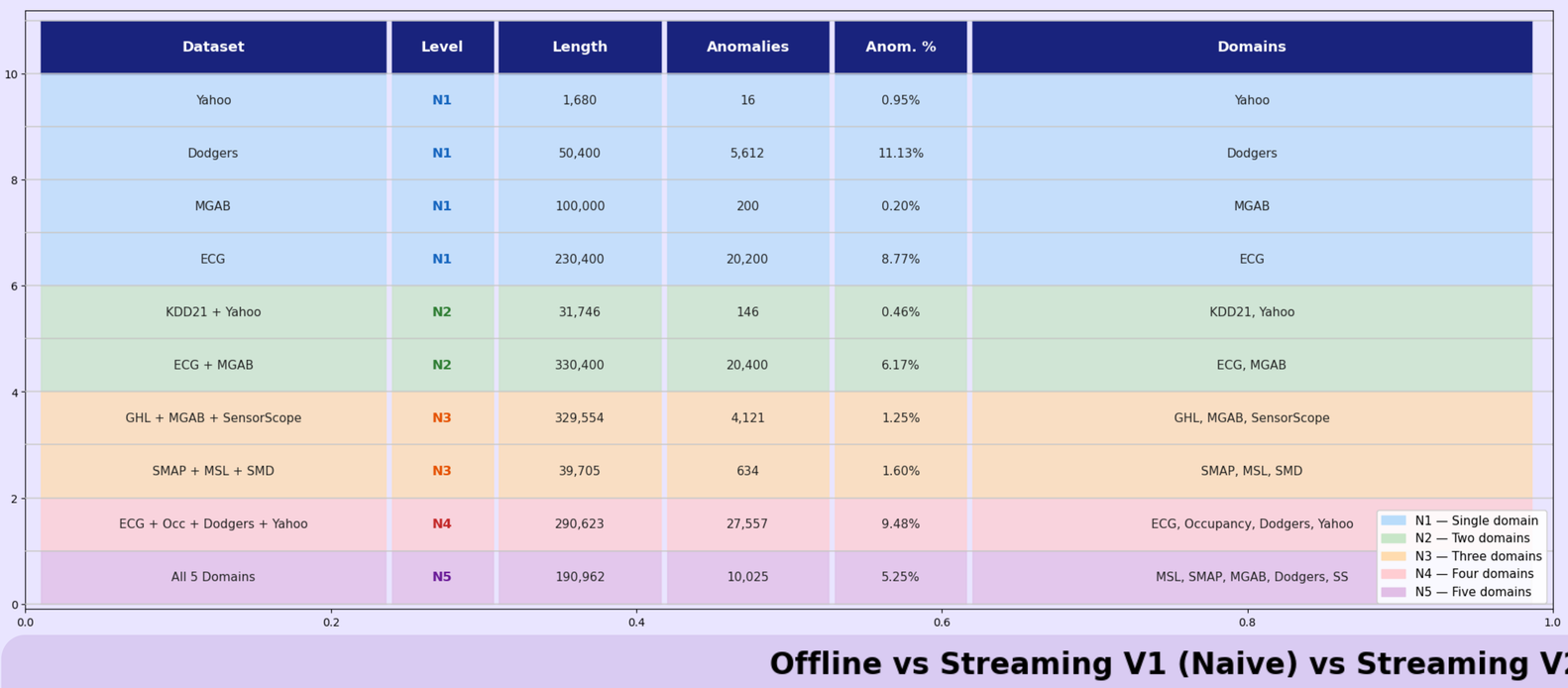
Introduction

Anomaly detection in time series is critical across many domains. Most methods that operate offline require the full dataset upfront, which is impractical when data arrives as a continuous stream from shifting distributions. **Goal**: **Adapt Isolation Forest, OCSVM, and LSTM to streaming settings** and evaluate how targeted modifications handle concept drift **benchmarked against SAND.**

Methods	Offline	Naive Streaming	Improved Streaming
Isolation Forest	Full-series training with sliding window	Retrained from scratch on every batch	**Drift-aware retraining** when EMA score exceeds threshold
OCSVM	Trained on the first 10% of the series	Applied independently to every batch	**KMeans** update when cluster structure shifts
LSTM	Trained on clean subsequences; reconstruction-error scoring	Retrained on each batch without memory	**Experience Replay** using current batch and stored normal samples
SAND	-	Online streaming baseline	-

Dataset Overview — TSB-UAD Benchmark

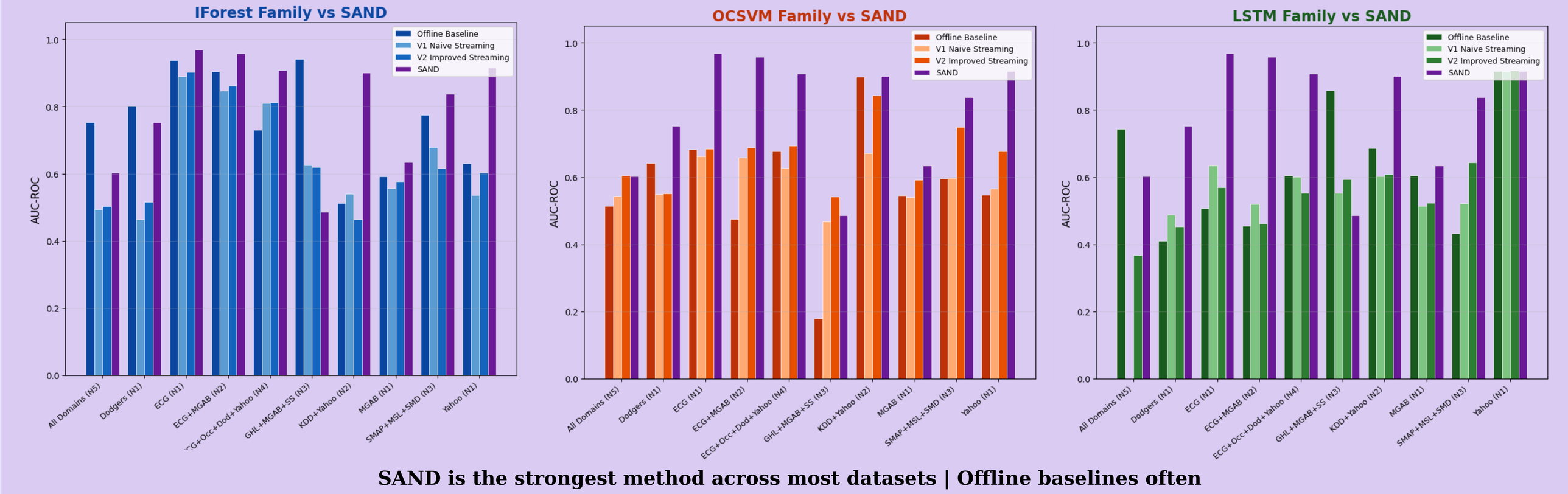
Dataset	Level	Length	Anomalies	Anom. %	Domains
Yahoo	N1	1,680	16	0.95%	Yahoo
Dodgers	N1	50,400	5,612	11.13%	Dodgers
MGAB	N1	100,000	200	0.20%	MGAB
ECG	N1	230,400	20,200	8.77%	ECG
KDD21 + Yahoo	N2	31,746	146	0.46%	KDD21, Yahoo
ECG + MGAB	N2	330,400	20,400	6.17%	ECG, MGAB
GHL + MGAB + SensorScope	N3	329,554	4,121	1.25%	GHL, MGAB, SensorScope
SMAP + MSL + SMD	N3	39,705	634	1.60%	SMAP, MSL, SMD
ECG + Occ + Dodgers + Yahoo	N4	290,623	27,557	9.48%	ECG, Occupancy, Dodgers, Yahoo
All 5 Domains	N5	190,962	10,025	5.25%	MSL, SMAP, MGAB, Dodgers, SS



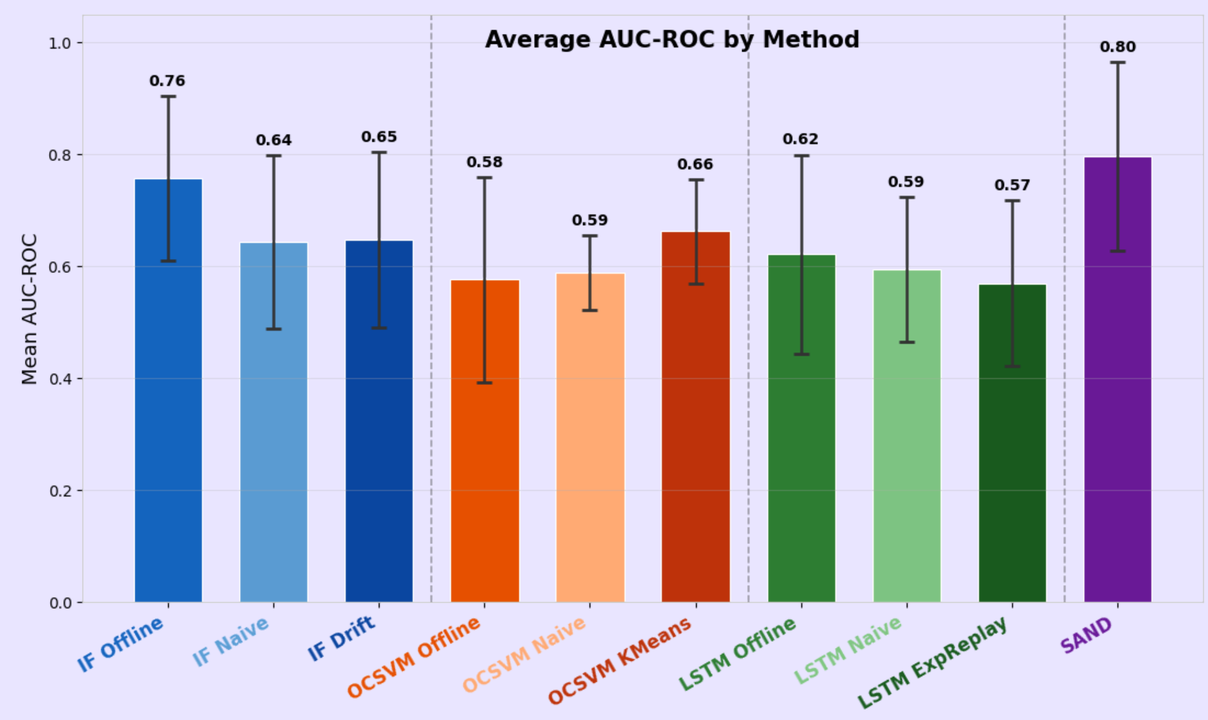
Experiments use 10 time series from **8 domains** (traffic, ECG, IoT, space telemetry, etc.)

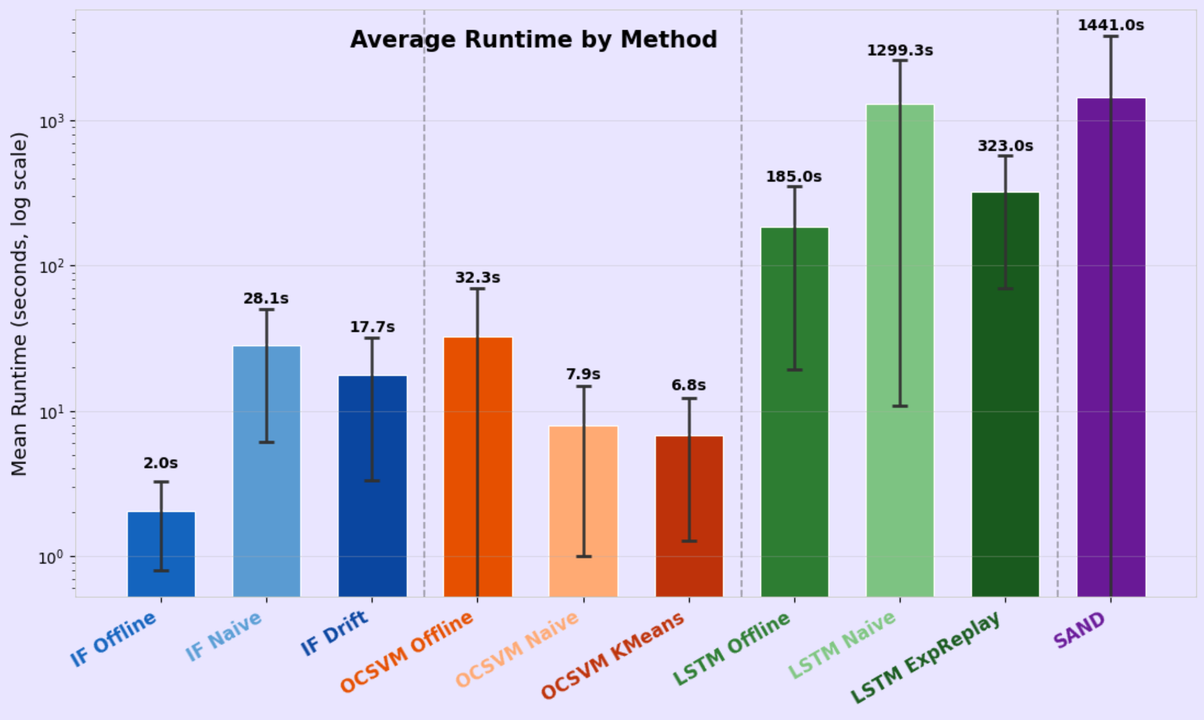
Series are organized into **5 normality levels**: N1 uses single-domain series| N2–N5 are constructed by concatenating series from 2–5 different domains, simulating progressive distribution shifts

Dataset **lengths** range from **1,680 to 330,400** time steps, with **anomaly rates** between **0.2% and 11%**



SAND is the strongest method across most datasets | Offline baselines often outperform their own streaming variants | OCSVM V2 consistently improves over V1





Takeaway: IF Offline offers the best speed-accuracy trade-off | OCSVM V2 achieves the largest V1→V2 improvement at low runtime cost | LSTM is slow without proportional accuracy gains